

第二章作业

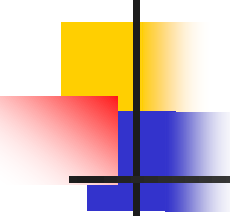
提交方式请按照文件“2026 电路与数字逻辑作业、课堂练习提交说明.pdf”的要求

请于2026年3月31日（周二 23: 59PM）前提交。

问题包括：

(1) 教材第80， 81页： 2.1(b,c) 2.2(b), 2.3(b), 2.4(a, c)

(2) T1, T2, T3, T4, T5, T6, T7



T1

Simplify the function into the simplest “AND-OR” form.

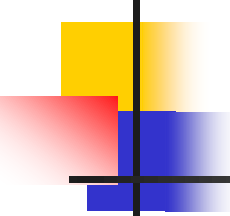
$$F = \overline{\overline{A}BC} + \overline{A}BCD + \overline{A}E$$

Simplify the function into the simplest “OR-AND” form.

$$F = (A + B + E) \cdot (A + \overline{B} + D) \cdot (A + D + E)$$

Simplify the following function into simplest AND-OR form by algebraic method.

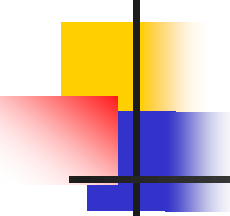
$$\overline{A} \overline{C} + \overline{A} D + \overline{A} C \overline{D} + A \overline{C} \overline{D} + ABD + A \overline{B}$$



T4 - Two Solutions

Simplify the function with K-map, and write the simplest “AND-OR” function.

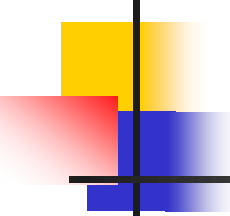
$$(a) F = \sum_{w,x,y,z} (0,1,3,5,14) + d(8,15)$$



T5

Simplify the function with K-map, and write the simplest “AND-OR” function.

$$(b)F = \sum_{w,x,y,z} (0,1,2,8,11) + d(3,9,15)$$



T6-Two Solutions

Simplify the function with K-map, and write the simplest “AND-OR” function.

$$(c)F = \sum_{A,B,C,D} (4, 6, 7, 9, 13) + d(12)$$

Simplify the function with K-map, and write the simplest “AND-OR” function.

$$(d)F = \sum_{A,B,C,D} (1, 5, 12, 13, 14, 15) + d(7, 9)$$